

7 Hidden Causes of Inverter Underperformance

Diagnostic checklist for phase imbalance, MPPT drift & efficiency loss

Purpose: Use this checklist quarterly or when inverter production falls below 90% of expected output. These issues cause 8-15% silent underperformance without SCADA alarms. Check each box after investigating and documenting findings. Keep completed checklists for warranty claims and performance trend analysis.

1 Phase Imbalance ☐

Symptom: One or two phases produce 5-12% less power than the third phase

Diagnostic:

- Compare 3-phase AC output during midday (11am-1pm) under clear sky conditions
- Calculate: $(\text{Max Phase} - \text{Min Phase}) / \text{Max Phase} \times 100\%$
- Normal: <3% difference | Warning: 3-8% | Critical: >8%

Action:

- Check AC contactor condition (contacts may be degraded on one phase)
- Verify string allocation is balanced across all three phases
- Inspect phase-specific cable connections and terminations

Impact: 8% phase imbalance = 2.7% total inverter capacity loss (~\$2,700/year per MW)

2

MPPT Drift



Symptom: DC voltage consistently 5-15V below optimal operating point

Diagnostic:

- Record DC voltage during steady-state operation (clear conditions, stable irradiance)
- Compare to expected V_{mp} from module datasheet adjusted for temperature:
$$V_{mp_actual} = V_{mp_stc} + (Temp - 25^{\circ}C) \times temp_coeff \times V_{mp_stc}$$
- Normal: $\pm 5V$ from expected | Warning: 5-10V | Critical: $>10V$ deviation

Action:

- Reset MPPT algorithm (power cycle inverter or initiate MPPT re-scan)
- Check for firmware updates addressing MPPT stability
- Verify string configuration matches inverter settings (number of modules per string)

Impact: 5% MPPT deviation = 3-5% power loss (~\$3,000-\$5,000/year per MW)

3

Component Degradation



Symptom: Gradual efficiency decline over months/years without obvious cause

Diagnostic:

- Calculate monthly DC-to-AC efficiency: $(AC \text{ Energy Out} / DC \text{ Energy In}) \times 100\%$
- Plot trend over 12+ months and calculate degradation rate (%/year)
- Normal: $<0.5\%/year$ | Warning: $0.5-1.5\%/year$ | Critical: $>1.5\%/year$

Action:

- Schedule thermal imaging inspection to identify hot spots (capacitor, IGBT degradation)
- Review maintenance logs for dust accumulation in cooling system
- Consider proactive component replacement if degradation accelerating

Impact: 1%/year degradation = \$10K cumulative loss over 10-year warranty period per MW

4

Reactive Power Mismanagement



Symptom: Inverter derates active power or fails grid compliance checks

Diagnostic:

- Compare commanded power factor (from SCADA setpoint) to actual (measured Q/P ratio)
- Calculate: Actual PF = $P / \sqrt{P^2 + Q^2}$ where P=active power, Q=reactive power
- Normal: <0.02 PF error | Warning: 0.02-0.05 | Critical: >0.05 or grid penalties

Action:

- Verify reactive power control mode (fixed PF, Q(U), $\cos \phi(P)$, etc.)
- Check for conflicting commands from SCADA vs local controller settings
- Ensure inverter apparent power rating not limiting active power delivery

Impact: 0.05 PF error can reduce active power by 5% + grid compliance penalties (\$500-\$2,000/incident)

5

Excessive Thermal Derating



Symptom: Power output drops below DC input capacity during high ambient temperatures

Diagnostic:

- Record ambient temperature, internal inverter temperature, and AC output power
- Compare to manufacturer derating curve (usually in datasheet or manual)
- Normal: Matches curve $\pm 5\%$ | Warning: 5-10% more aggressive | Critical: >10% below expected

Action:

- Check cooling system: fans operational, air filters clean, ventilation unobstructed
- Review derating settings in inverter configuration (may be set too conservatively)
- Consider external shading or improved ventilation for inverters in hot locations

Impact: 10% excess derating during peak hours = 3-5% daily energy loss in summer months (~\$3,000-\$5,000/year per MW)

6 String-Level Mismatch ☐

Symptom: Inverter consistently underperforms vs identical neighbors with same irradiance

Diagnostic:

- Compare inverter output to 3-5 identical neighbors (same model, string config, exposure)
- Calculate: $(\text{Target Inverter} / \text{Avg of Neighbors}) \times 100\%$
- Normal: 95-105% | Warning: 90-95% | Critical: <90%

Action:

- Inspect DC combiner box connections and fuses for one weak string
- Check individual string open-circuit voltage (Voc) to identify underperforming string
- Investigate shading, soiling, or module failures on specific string

Impact: One string 15% underperforming = 5-8% total inverter loss if 8-12 strings (~\$5,000-\$8,000/year per MW)

7 Intermittent Communication Loss ☐

Symptom: Missing data points during peak production hours, daily totals seem low

Diagnostic:

- Review SCADA data completeness: count missing timestamps during daylight hours
- Correlate data gaps with peak irradiance times (10am-2pm most critical)
- Normal: >99% complete | Warning: 95-99% | Critical: <95% or pattern of peak-hour gaps

Action:

- Check communication cable quality (RS485/Ethernet) for noise or damage
- Verify network topology: avoid daisy-chain lengths >1km without repeaters
- Review SCADA polling frequency and buffer capacity (may be saturated during high data periods)

Impact: 5% data loss during peak hours = 3-4% underestimation of production + false performance issues (~\$3,000-\$4,000/year per MW)

Need Help Diagnosing Your Inverters?

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